Gallium

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Ga

Similar in appearance to aluminum, gallium is a soft, silvery metal. However, unlike aluminum, this element has a much lower melting point of 84.2°F (29°C)—a cube of pure gallium will melt in your hand due to body heat. Some compounds of gallium are useful in semiconductors and are common in the electronics industry.

Diaspore

Solid piece of pure gallium melting

Diaspore

Minerals that contain gallium are rare, but very small amounts of it can be found in ores such as the aluminum oxide mineral diaspore. Trace amounts of gallium are present in some other ores, including sphalerite, germanite, and bauxite.

Rover Zoë in the Atacama Desert

Searching for life

Zoë is a rover built at Carnegie Mellon University, in Pennsylvania. Its solar panels generate power using cells made of a compound called gallium arsenide. In 2005, this rover was used for the first time in South America's dry Atacama Desert to search for signs of microscopic life. It returned in 2013 with a long drill, to look beneath the desert's surface. The goal is to develop technologies to explore Mars.

Array of colors

Gallium arsenide is one of the gallium compounds that can convert electricity to light in LEDs (light-emitting diodes). Available in many colors, LEDs are up to 80 percent more energy-efficient than traditional light bulbs and have many uses—from lights in homes to traffic lights and vehicle lights.